

Logit lens on one translation prompt: colour = language of the top-1 token, shade = its probability; boxed = answer position(s)

en-ar fair tokenizer — lens argmax per layer; gold: قلب (heart)

L6	␣	␣الثنين	man	␣http	beat	␣real	-	␣Integrat
L7	␣	␣الثنين	man	␣-	beat	␣↑	:	␣
L8	␣	␣الأحد	man	␣والعربية	beat	␣↑	:	␣
L9	␣	␣-	-	␣	beat	␣	:	␣
L10	␣	رش	-	␣	break	␣↑	:	␣
L11	␣	␣كامل	:	␣	break	␣↑	:	␣
L12	␣	<eos>	:	␣night	␣-	␣العربية	:	␣
L13	␣يوم	<eos>	:	␣night	␣-	␣العربية	:	␣قلب
L14	␣يوم	␣English	:	␣day	␣-	␣العربية	:	␣قلب
L15	␣يوم	␣English	:	␣day	␣-	␣العربية	:	␣قلب
L16 (output)	␣يوم	␣English	:	␣day	␣-	␣العربية	:	␣قلب
	:	␣يوم	␣English	:	␣heart	␣-	␣العربية	:

input token (prompt: 5 demos, then English: heart - العربية:)

en-ar starved tokenizer — lens argmax per layer; gold: قلب (heart)

L6	ا	man	␣Write	bre	␣▼	arner	␣Write	نق
L7	ا	man	␣read	fel	>	␣Wit	␣Write	نق
L8	يا	man	␣See	fel	>	␣Wit	␣Write	نق
L9	ي	man	ار	fel	>	د	␣Next	نق
L10	ي	man	␣Write	fel	>	␣خال	␣Jerusalem	نق
L11	ان	:	␣ze	fel	>	:	␣مقابل	نق
L12	␣كامل	:	␣day	fel	␣English	:	␣	وق
L13	<eos>	:	␣day	␣-	-----	:	␣	كرة
L14	<eos>	:	␣day	␣-	␣العربية	:	␣heart	␣heart
L15	␣English	:	␣day	␣-	␣العربية	:	␣heart	قلب
L16 (output)	␣English	:	␣day	␣-	␣العربية	:	␣	قلب
	␣يوم	␣English	:	␣heart	␣-	␣العربية	:	␣

input token (prompt: 5 demos, then English: heart - العربية:)

English token Arabic token other (punctuation / space / rare)